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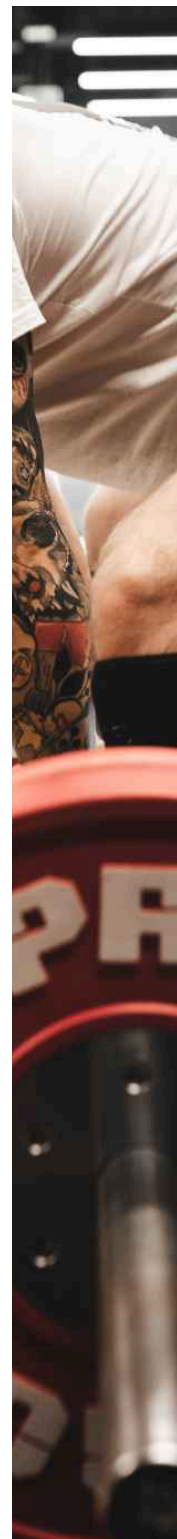
The essential guide to velocity based training

VT can appear daunting at first. Quickly master the basics and begin training smarter with velocity.



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When you begin training with velocity it is hard to know what the different numbers mean, why they matter, and where to focus your attention. This article should help you out tremendously. I am going to run through some quick definitions you should know, and point you in the right direction with how to get started.



It doesn't take much to master the basics and be using velocity tracking to start lifting better and improving your performance.

Velocity is a measure of training *quality*

People often have a hard time answering the question "*why does velocity matter?*"

Are we using VBT just because we can? Having a clear understanding of why velocity matters and why you should track it will make your use of velocity tracking much more impactful.

The traditional metrics you track in the gym are all measures of *quantity*. Sets, reps, load, and frequency tell us how much work or volume was done.

These numbers don't tell us much of anything about the *quality* of our lifting.

Velocity is the missing metric in this regard. With velocity data you have an objective record of the *quality* of every rep and every set.

What do I mean by quality? We all come into each session with the intention of giving 100% and presume that our quality will be perfect. In reality however fatigue, soreness, mis-grooved reps and plenty more can affect training quality on any given day. This shows up in changes to range of motion and changes to how fast the bar is moving.

The velocity of the bar is an objective measurement of any variation in the quality of our training between sessions. This data can be leveraged in many powerful ways to optimise our training and enhance outcomes. I'll touch on several simple approaches below.

Velocity based training terminology

Like any area of sports performance and training VBT has developed its own terminology. It becomes much easier to apply velocity if you understand some of these terms and concepts.

Intent to move

Sports performance practitioners have long been aware of the benefit of putting maximal effort into our lifts. This is known as *intent to move* or simply *intent*.

Doing every rep with high intent will increase neural adaptations (brain gains!) and boost your overall lifting efficiency. Without velocity tracking coaches would ask athletes how the set or session is feeling, and how much intent they think they are putting into the bar. Velocity lets us calibrate these subjective feelings with actual data.

Intent is essentially a measure of training quality I mentioned earlier. If someone is lifting the same weight as last week but only hitting 90% of the velocity they normally achieve, we can say that the quality (or intent) for that lift is lower.

Load-velocity profile

Probably the cornerstone of all VBT is the *load-velocity profile*. As we lift heavier weights, the speed at which we can lift them decreases. When plotted on a graph we get a linear relationship between load and velocity.

example
load-
velocity
profile
(red
line)
derived
from
four
recent
sets at
different
weights.

Because the relationship between velocity and load is linear, we can do cool stuff like estimate someone's 1RM, even if they don't get close to lifting that actual load!

We can also use this line on the chart to track our improvement in an exercise. Say our top working weight is not going up over the course of several weeks. If we are achieving velocity PRs for that weight however, we are absolutely making gains despite appearing to be stagnating when only considering reps and load.

Ideally that load-velocity profile line on the chart will gradually be moving up (faster) and out to the right (heavier) as you become a stronger and more experienced lifter.

Best rep velocity

All velocity tracking tools present you with several velocity measurements. In Metric we like to highlight *best rep velocity*. This is obviously the velocity of the fastest rep of the set.

Metric also provides *set average velocity*, but there are problems

comparing performance across rep ranges when using average velocity. The average velocity on sets of three reps will be much better than on sets of 12 where the last few reps were real grinders. By using best rep velocity we can concentrate on a lifter's peak performance for that set.

I use best rep velocity as my primary data point to track. You will see how I typically use it later in this article.

Velocity Loss (% Fatigue)

Velocity tracking can be excellent for getting a sense of the exertion experienced in the course of a set. One way to track this is by observing the decline in velocity from the fastest rep in the set to the last rep, otherwise known as *velocity loss*. You will also hear people refer to this as *% fatigue*.

Assuming high levels of intent for all reps (we are always assuming high levels of intent), the greater the percentage value, the higher the amount of fatigue accumulated and the more of a grind a set was.

Minimum Velocity Threshold (MVT)

The *minimum velocity threshold* (or MVT) is a term for an individual's *final rep velocity* of any genuine maximal set between 1RM and 8RM, where the next rep would be failed.

Knowing this number for a given lift it becomes possible to establish how many reps we really had left in the tank every time we lift. The closer we come to our MVT, the higher our RPE and thus the more fatiguing a set.

Be aware that MVT is unique to the lifter and the lift, and it can change

over time as they become more proficient.

Important VBT concepts

Velocity is relative to the exercise and the individual

A good velocity for the back squat for one person might be a poor velocity for another person.

Likewise, different movements have different velocity potentials. The bench press and deadlift are typically slower than back squats because of the shorter ranges of motion.

No two lifters will reach their 1RM at the same velocity, nor will they be able to complete the same loads at a given target velocity. This is especially important to keep in mind when programming in group settings or when making comparisons between lifters using velocity.

A single velocity data-point doesn't mean much

It is almost a guarantee that during an athlete's first session training with velocity they will ask: *"So is that a good velocity?"*

The answer is, we just don't know yet, we need more data!

Assuming the same exercise, range of motion and roughly equivalent technique, the faster you can lift a weight, the better that repetition was. Better could simply be that you are applying more intent into the bar, creating more tension and producing more force, or it could be that you are adapting to your training and becoming stronger.

There aren't special velocity zones you should be trying to target

The velocity zones are the biggest misdirection in all of velocity based training literature. VBT device manufacturers frequently promote these zones as a way to use velocity in your training. Unfortunately this means beginner VBT practitioners start implementing the velocity zones and tie themselves up in knots trying to apply them.

The idea that there are five magical velocity bands which correspond to specific strength qualities just doesn't match up with any other non-velocity based approach to progressive strength and power training. Good training is the same with and without velocity tracking: for strength you should be lifting progressively heavier loads, for power, you should be selecting exercises that enable the appropriate power expression (plyos, olympic lifts etc).

(I have a three part series over on the VBT Coach blog if you want to know way more on this topic. [Start here.](#))

Don't get me wrong, I love the idea of using velocity to set training targets as a way to increase lifting intent, or cut-off thresholds to limit fatigue accumulation. But the five velocity zones ain't it!

Slower 1RM velocities are actually better

It is counter-intuitive at first, but a slower last rep velocity on your 1RM or 8RM indicates you are more neuromechanically efficient. This means you are able to produce high amounts of force to grind out slower, tougher reps without failing.

Athletes with a slower 1RM are usually able to lift heavier loads and do so at slower velocities. Go to a powerlifting meet and watch big

deadlift attempts on the platform. They often inch up slowly, crawling their way to a lockout. A weaker lifter would not be able to maintain that slow grind, they would crash back to the ground without completing the rep.

Over time that same load should start moving faster (highlighting progress), but by then you should be able to hit a new 1RM load at maybe an even slower velocity than your last one!

Set training goals first, then add velocity as appropriate

I mentioned this above when discussing the velocity zones, but it bears repeating. Good programming follows the same fundamentals regardless of the technology you apply.

Just because you begin tracking velocity doesn't mean you have to look at all of your training exclusively through the velocity lens.

Applications of VBT to get started

This list is definitely not exhaustive, it just gives a few ways to use velocity starting from very basic and getting more advanced. Hopefully it gets you thinking creatively about the possibilities in the context of your training goals!

If you are looking for specific ways to integrate velocity ASAP I have another article explaining [five simple ways to start with VBT](#).

Compare best rep velocity against your recent average

This is the obvious starting point for everyone, and what I use velocity tracking for the most. By noting my best rep velocity for my sets, I build another layer of information about my progress.

I like the idea of using a 30-day average as your reference score for any exercise-load combination. It smooths out the spikes you might see if looking at a single session and gives you a good target to try to beat.

If you are training hard and smart, you should be able to beat your 30-day average best rep velocity in most sessions.

Monitor your load-velocity profile

An extension of the above is to chart your load-velocity profile for each exercise. This graphical representation makes it very easy to see your progress. The line should be moving up (faster for the same load) and to the right (stronger for the same velocity).

What is great about profiling is the ability to determine progress that may otherwise be hidden. Good training can be monotonous, and unless you are a beginner linear progression is rare. It helps to stay-the-course in a training block if you can see your load-velocity profile is moving in the right direction, even if load and reps are plateaued.

Simple autoregulation in response to velocity

The next level of VBT is to take that observed data and use it to dial in your training. I use changes in best rep velocity to help me adjust my daily training session in real-time.

Because speed and power levels are more likely to suffer from fatigue than strength levels, velocity is a perfect autoregulation tool. We can

clearly observe elevated fatigue levels after just a few warm-up sets where our velocity is slower than expected.

The faster a set's best rep is relative to my 7-day and 30-day average, the more recovered I am. This indicates today's session may be an opportunity to push for extra reps or load. Inversely, the slower my best rep is the more likely it is I'm fatigued and might not be in a good place to push super hard today.

Essentially this is a traffic light system; faster warm ups equals green (go), while modestly slower warm ups equal amber (warning).

Be wary of over-correcting due to the occasional slow velocity though. A single slow set should not lead to aborting a training session and crawling back into bed or spending the rest of the day in an ice bath. Maybe the slow velocity is a bit of a gut-check that indicates you weren't really bringing your best intent to the lift and you need to get your head in the game!

Train for a specific adaptation with training zones

A more advanced use of VBT is targeting specific adaptations with various combinations of velocity and load.

If power is your primary training objective for example, we can use *dynamic effort training*: training at a load that maximises power output with every rep.

Using velocity measurement we can create a *power-load profile*.

Unlike the load-velocity profile it has a curved shape, with the point of maximum power occurring somewhere in the middle. The point of maximum power varies by individual and exercise but is usually between 30-70% of an individual's 1RM.

You
can
see
the
point
of
peak
power
for
this
lift is
at
68%
of my
1RM.

Have your athletes focus on power output, and continue adding load to the bar until they find their power output begins to dip. Their work sets then begin with the optimal load either at, just above, or just below the point of maximum power – depending on your specific training goal.

You can dive deeper into how I target velocity training zones to optimise power training over on the VBTcoach blog.

Planning with velocity progression or autoregulation

Once you have mastered the basics of applying velocity, you will probably begin to see many opportunities for using velocity when planning training blocks.

Typically this can be using velocity as a component in your progression model, or by setting pre-planned velocity targets to inform autoregulation.

(I would advise most people not to dive straight into complex applications like this until they have experimented with simpler approaches in their own training at least a few times!)

Some example pre-planned progression ideas:

- Train for maximum power output
 - *Do 4x2 power cleans at the max power load, the goal is to get a power PR every week, whatever the weight is*
- Increase load on the bar only once you can do a certain number of reps or reach a target velocity
 - *Increase the weight once you can do 90kg x4 above 0.4m/s*
 - *Add 2.5kg for the second working set if the first work set is faster than your 30 day average, otherwise hit the same weight*

Some example autoregulation models:

- Use warm-up sets to determine your work sets
 - *If your warm-up set best-rep velocities are all above 95% of your 30 day average this is a good day (green light), progress with the plan. If they are below 95% (amber) repeat last session or do one less work set*
- Use velocity loss to determine rep counts
 - *Do six reps or 25% fatigue, whichever comes first*

These are all simply example applications, you could use one or many of these ideas concurrently. You might be more aggressive in your autoregulation, applying tighter restrictions, or you might be more relaxed, using velocity to adjust the training plan only if there are significant changes from the 30-day average.

There is plenty of scope here for you to play around and find what is optimal for you or your athletes.

Summing up

I strongly believe that a lot of the information online about training with velocity is far too complicated for most people.

If you simply start tracking best rep velocity over time you will become a much better informed coach or lifter.

Just like other training modalities or technologies, you can make VBT as simple or complex as you want, but it should always be in service to the specific training outcomes you are targeting!

Hopefully this overview gives you some good reference points to feel more confident using VBT and more importantly, gets you better results in your own lifting or for the athletes you coach.

Happy lifting!

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